

5054_ALL_10marks

Total Marks: 10

- 34 A step-up transformer is connected between a power station and a long-distance electricity transmission cable.

What is the purpose of the step-up transformer, and does it operate using alternating current (a.c.) or using direct current (d.c.)?

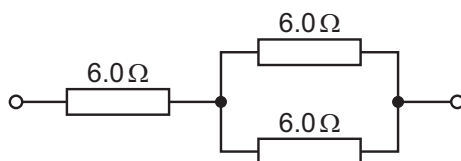
	purpose of a step-up transformer	current used
A	to decrease the current in the cable	d.c.
B	to decrease the current in the cable	a.c.
C	to increase the current in the cable	d.c.
D	to increase the current in the cable	a.c.

- 16 When a liquid evaporates, molecules escape from its surface.

Which molecules escape, and what happens to the average kinetic energy of the molecules remaining in the liquid?

- A The less energetic molecules escape and the average kinetic energy decreases.
- B The less energetic molecules escape and the average kinetic energy increases.
- C The more energetic molecules escape and the average kinetic energy decreases.
- D The more energetic molecules escape and the average kinetic energy increases.
- 3 Which statement explains why a heavy coin that falls through a short distance towards the ground does **not** reach terminal velocity?
- A The coin has **not** hit the ground.
- B The weight of the coin equals the air resistance.
- C The weight of the coin increases as the air resistance increases.
- D The weight of the coin is always more than the air resistance.

- 31 Each of the resistors in the circuit shown has a resistance of 6.0Ω .



What is the total resistance of the circuit?

- A 2.0Ω B 4.0Ω C 9.0Ω D 18Ω

18 Thermal energy passes quickly from hot regions of a metal to cold regions of the metal.

Why are metals good conductors of thermal energy?

- A** Electrons in the metal vibrate about fixed points and quickly pass energy from one electron to the next.
 - B** Electrons move quickly from ions in hot regions to ions in cold regions.
 - C** Protons in the metal are close together and move quickly from hot to cold regions.
 - D** Small ions of the metal move quickly from hot to cold regions through spaces in the metal lattice.
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28 A 12 V battery is connected to a resistor of resistance $100\ \Omega$.

How much charge passes through the resistor in 30 minutes?

- A** 3.6 C **B** 220 C **C** 250 C **D** 15 000 C
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15 A student determines the specific heat capacity of a metal by carrying out an experiment.

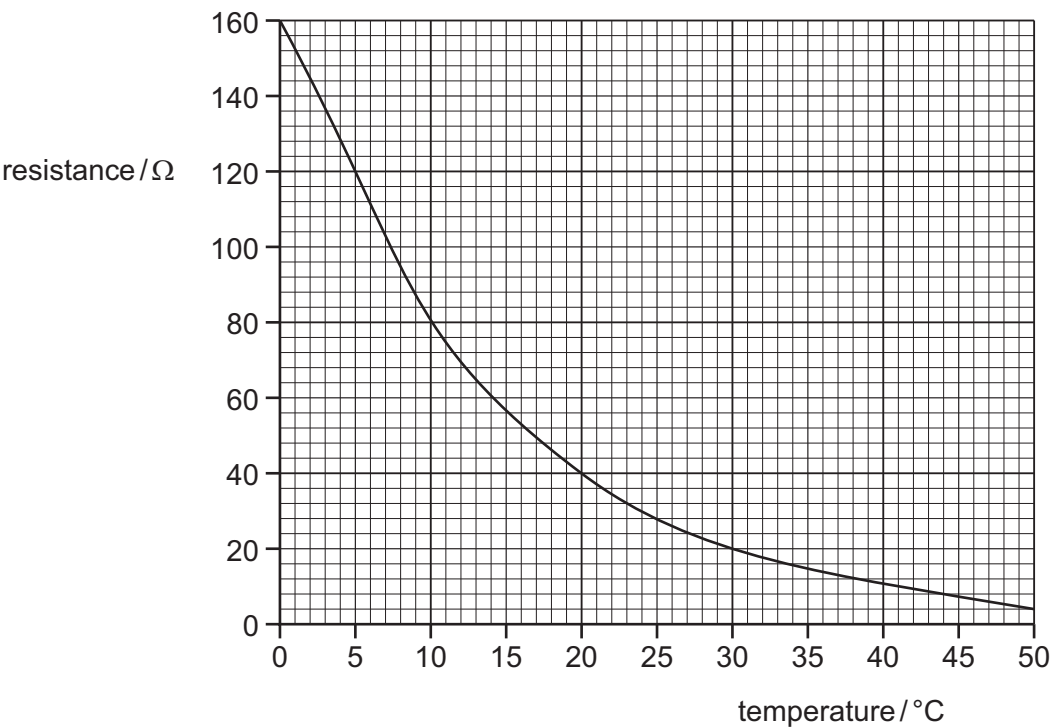
The list shows the apparatus that the student has available.

- 1 electronic balance
- 2 ruler
- 3 thermometer
- 4 heater operating at a power of 50 W
- 5 stopwatch
- 6 ammeter
- 7 voltmeter

Which apparatus does the student use?

- A** 1, 3, 4 and 5
 - B** 1, 3, 6 and 7
 - C** 2, 3, 4 and 6
 - D** 3, 4, 5, 6 and 7
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30 The graph shows how the resistance of a thermistor varies with temperature.



The thermistor is at a temperature of 10 °C and is connected in series with a fixed resistor of resistance 20 Ω.

What is the combined resistance of both components?

- A** 16 Ω **B** 20 Ω **C** 80 Ω **D** 100 Ω

29 A circuit consists of a resistor connected between the positive terminal and the negative terminal of a power supply.

Which row describes the direction of conventional current and the direction of flow of free electrons in the resistor?

	direction of conventional current	direction of flow of free electrons
A	from negative to positive	from negative to positive
B	from negative to positive	from positive to negative
C	from positive to negative	from negative to positive
D	from positive to negative	from positive to negative

10 The mass of object P is greater than the mass of object Q.

The objects contain different amounts of matter and have a different resistance to change of motion.

Which row is correct?

	greater amount of matter	greater resistance to change of motion
A	P	P
B	P	Q
C	Q	P
D	Q	Q
